

Humans often make poor judgments due to cognitive biases, fatigue, incorrect prior beliefs, or inattentiveness. While AI systems have the potential to support these judgments, humans frequently struggle to collaborate with AI effectively because the same underlying limitations hinder their interaction.

My research develops foundational frameworks for understanding and improving human-AI collaboration. To date, I have demonstrated that common assumptions about AI are flawed – explanations fail as safeguards against algorithmic discrimination, and interpretability increases overreliance on automated systems. Conversely, I have also demonstrated productive pathways forward by developing methods for interpreting reinforcement learning (RL) policies and showing that they can be applied to genuinely improve human decision-making.

Building on these insights, I outline an agenda for two projects: (1) studying user behavior with generative AI systems to understand what interface designs work best, and (2) creating systems that learn from user feedback to personalize AI outputs more efficiently. Together, these projects bridge the gap between AI capabilities and user needs.

To date, my work has fostered international collaborations across machine learning, cognitive science, philosophy, and public policy, with publications at top venues including Machine Learning, Behavior Research Methods, Cognitive Science, and FAccT.

Research Areas: Human-Centered AI, AI for Ethics & Fairness, Interactive ML & Agents

Proposed Research

Area: Human-Centered AI

The gap between AI model performance in controlled evaluations and practical utility in real-world deployment represents a fundamental challenge limiting the impact of current AI systems. Closing this gap requires both (1) better methods for adapting AI to user needs and (2) a deeper understanding of how users interact with these systems. In what follows, I describe two projects that advance these directions.

Specializing LLMs Without Fine-Tuning

Language models built for specialization and personalization represent a natural evolution in AI development. Such systems enable efficient code generation suited for specific teams (deployed across companies like Google, Microsoft or IBM) and can democratize access to therapeutic or medical support. However, obtaining specialized models currently relies on computationally expensive re-training or fine-tuning. Recently, researchers proposed to use rubrics – linear scoring functions on features that represent meaningful evaluation criteria (like ‘conciseness,’ ‘doc strings,’ or ‘comments’ in code generation) – to score model responses and select optimal outputs. Rubrics have demonstrated success code evaluation and medical education, but their effectiveness depends heavily on rubric weights, which are currently set by experts using heuristics and common sense. As a result, this process is consuming a lot of time and resources.

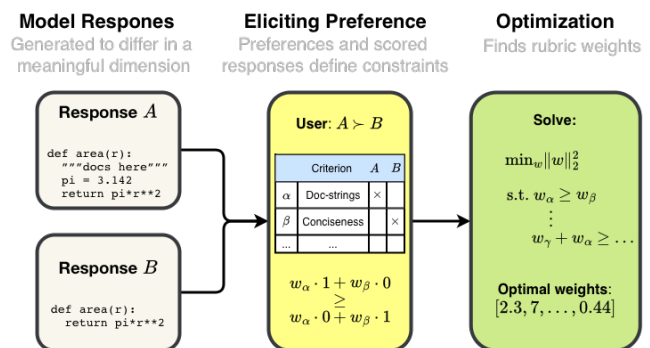


Fig. 1: Converting user preferences into optimized rubric weights.

In this project, I will automate the process of rubric creation by learning optimal weights from user preferences (see Fig. 1). My approach will associate each rubric criterion $c: R \rightarrow \{0, 1\}$, where $c(r)$ indicates whether response r satisfies criterion c , as a unit vector v_c . I will gather user preferences on pairs of model responses differing in c : $r_c \succeq r'_c$ where $c(r_c) = 1$ and $c(r'_c) = 0$. Each model response r can be then represented as a binary vector $V_r = \sum_{\{c: c(r)=1\}} v_c$. This representation allows user preferences to be encoded as linear inequalities: if $r_c \succeq r'_c$ then $w \cdot V_{r_c} \geq w \cdot V_{r'_c}$ where w represent the rubric’s weights, and use these inequalities to solve an optimization problem. This framework extends naturally to multi-stakeholder alignment. When preference conflicts arise, we could employ selective aggregation and establish partial orderings where preference exists only when it has enough support in the data.

My hypothesis in this project is that automated rubric weight optimization will produce more accurate and consistent evaluations compared to expert-defined heuristic rubrics. Since optimization offers a principled, data-driven approach to rubric creation, this research can significantly speed up rubric generation and offer a scalable solution for creating specialized and personalized models.